SURF SNELLIUS · ACCESS GUIDE

Connecting to Snellius

SSH keys on Windows, macOS and Linux

Official documentation: Connecting to the system

OVERVIEW

Five steps to log in

1	**Check OpenSSH**	Install the client if it is missing
2	**Create a key**	ed25519, protected by a passphrase
3	**Load the agent**	Unlock the key once per session
4	**Register it**	Paste the public key into the SURFcua portal
5	**Connect**	From a terminal or from VS Code

TERMS

Two ideas to know

Key pair

A private key that never leaves your machine, and a public key that you give to Snellius.

ssh-agent

A background program that holds your unlocked key, so you type the passphrase once. The "passkey manager".

Only the .pub file is ever shared.

TERMS

Placeholders in this guide

Placeholder	Meaning	Example
<user>	Your Snellius username	jdoe
<key_suffix>	A short label for your key	snellius
<name>	Host nickname in your SSH config	snellius
<windows_username>	Your Windows login, usually different from <user>	jane

Key files: id_ed25519_<key_suffix> (private) and id_ed25519_<key_suffix>.pub (public).

STEP 1 · SKIP IF YOU ALREADY HAVE IT

Install OpenSSH

Windows

Admin PowerShell

```
Add-WindowsCapability -Online -Name OpenSSH.Client~~~~0.0.1.0
```

Or Settings → System → Optional features → OpenSSH Client. Restart PowerShell, then run ssh -V.

macOS

```
brew install openssh
```

OpenSSH ships with macOS, so this is rarely needed. Homebrew gives a newer version.

Linux

```
sudo apt install openssh-client  # Debian, Ubuntu
sudo dnf install openssh-clients # Fedora, RHEL
sudo pacman -S openssh               # Arch
```

Then run ssh -V again. It should print a version.

STEP 1 · CHECK IT WORKED

Check OpenSSH

Windows

```
ssh -V
```

Should print a version. If it fails, repeat the previous slide.

macOS

```
ssh -V
```

Preinstalled. Verify in Terminal.

Linux

```
ssh -V
```

Usually preinstalled. Should print a version. If it fails, repeat the previous slide.

STEP 1 · INSTALL THE AGENT

Get an agent

Windows

```
Get-Service ssh-agent
Add-WindowsCapability -Online -Name OpenSSH.Client~~~~0.0.1.0 # only if not found
```

Often there already, or installed with the OpenSSH Client (step 1). You enable it in step 3.

macOS

```
ssh-add -l
eval "$(ssh-agent -s)" # only if it cannot connect
```

Built in and started for you. Nothing to install.

Linux

```
sudo apt install openssh-client # Debian, Ubuntu
sudo dnf install openssh-clients # Fedora, RHEL
eval "$(ssh-agent -s)" # headless machines
```

Often there already, or installed with the OpenSSH client.

STEP 2

Create the key

Windows

PowerShell

```
ssh-keygen -t ed25519 -f $env:USERPROFILE\.ssh\id_ed25519_<key_suffix> -C "<user>-snellius"
```

macOS / Linux

Terminal

```
ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519_<key_suffix> -C "<user>-snellius"
```

Enter a strong passphrase when asked.

This creates a private key (keep it secret) and a .pub file (the one you register).

STEP 3 · VERIFY WITH SSH-ADD -L

Add the key to the agent

Windows

```
# Admin PowerShell, once
Set-Service ssh-agent -StartupType Automatic
Start-Service ssh-agent
# Then, in a normal PowerShell
ssh-add $env:USERPROFILE\.ssh\id_ed25519_<key_suffix>
```

macOS

```
ssh-add --apple-use-keychain ~/.ssh/id_ed25519_<key_suffix>
```

Linux

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519_<key_suffix>
```

STEP 4

Get your public key

Windows

```
cat $env:USERPROFILE\.ssh\id_ed25519_<key_suffix>.pub
```

macOS / Linux

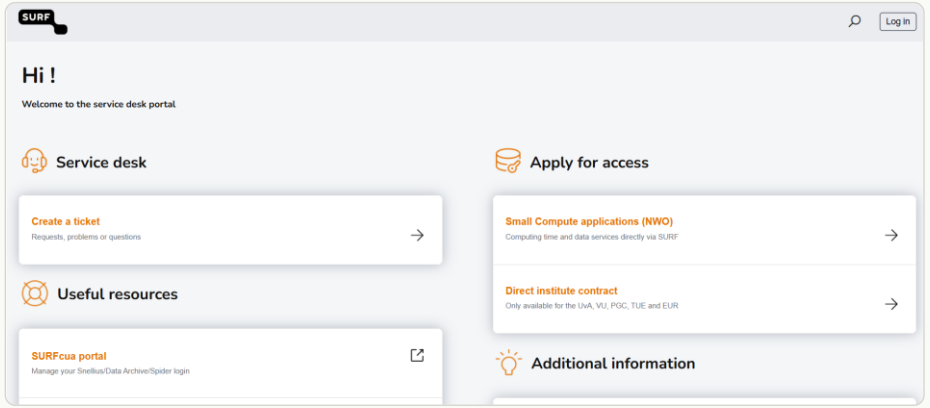
```
cat ~/.ssh/id_ed25519_<key_suffix>.pub
```

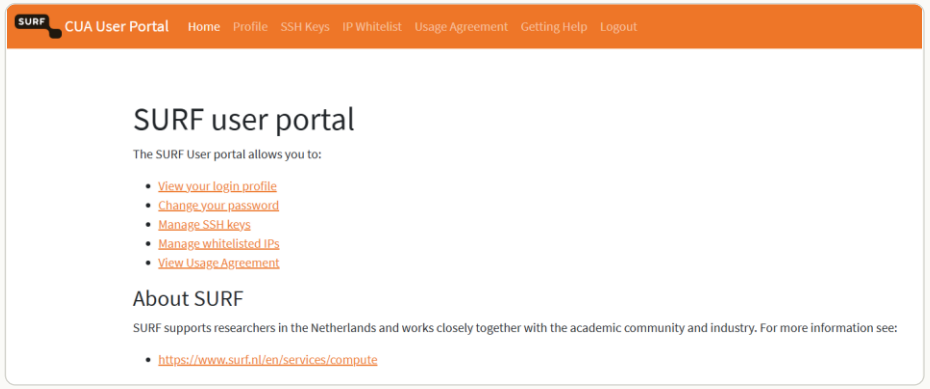
Copy the whole line, from ssh-ed25519 to the comment.

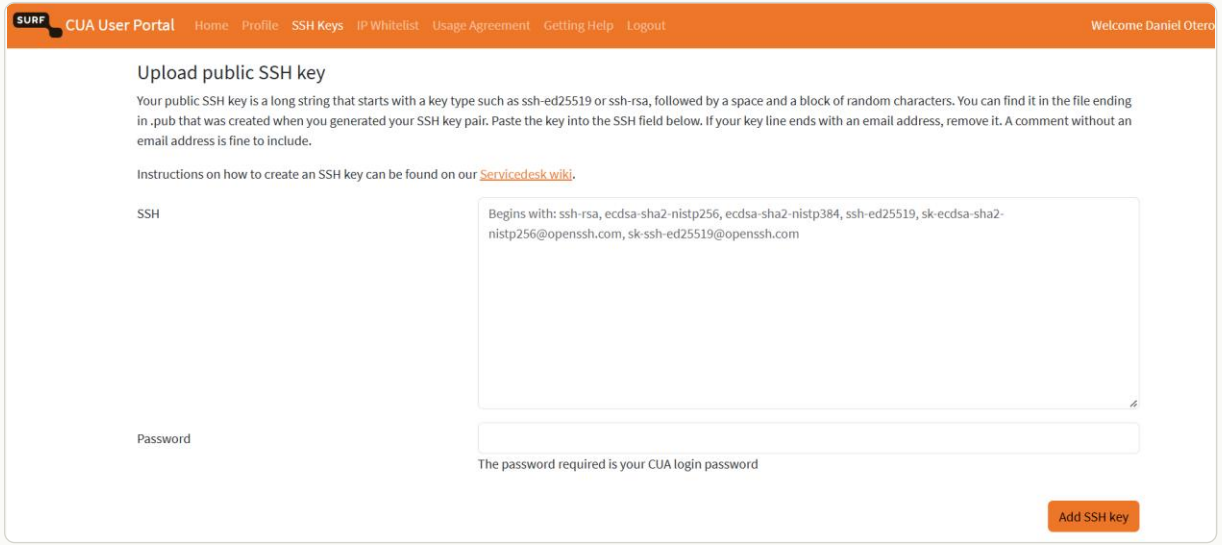
Next: paste it into the SURFcua portal.

STEP 5 · SCREENSHOTS

SURF portal screenshots







1 Service Desk: open the SURFcua portal

2 SURFcua portal: open SSH Keys

3 Paste key, enter CUA password, Add SSH key

STEP 6 · WINDOWS

Add the host to your SSH config

```
Host <name>
  HostName snellius.surf.nl
  User <user>
  IdentityFile C:\Users\<windows_username>\.ssh\id_ed25519_<key_suffix>
  AddKeysToAgent yes
  ForwardAgent yes
```

The file is C:\Users\<windows_username>\.ssh\config, with no extension.

IdentityFile points to the private key, including the suffix.

ForwardAgent yes lets Snellius use your local keys, such as GitHub.

STEP 6 · MACOS AND LINUX

Add the host to your SSH config

```
Host <name>
  HostName snellius.surf.nl
  User <user>
  IdentityFile ~/.ssh/id_ed25519_<key_suffix>
  AddKeysToAgent yes
  ForwardAgent yes
  UseKeychain yes # macOS only
```

The file is ~/.ssh/config.

Remove the UseKeychain line on Linux, or SSH reports an unknown option.

IdentityFile points to the private key, including the suffix.

ForwardAgent yes lets Snellius use your local keys, such as GitHub.

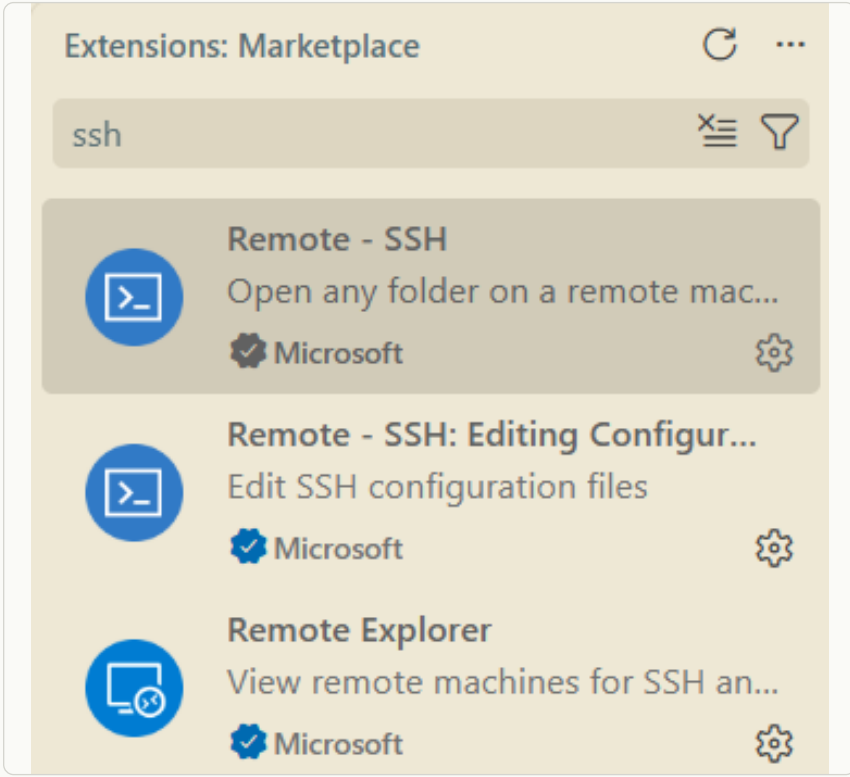
STEP 7 · VS CODE · 1 OF 2

Install the Remote - SSH extension

Open Extensions in VS Code.

Search for **ssh**.

Install **Remote - SSH**.



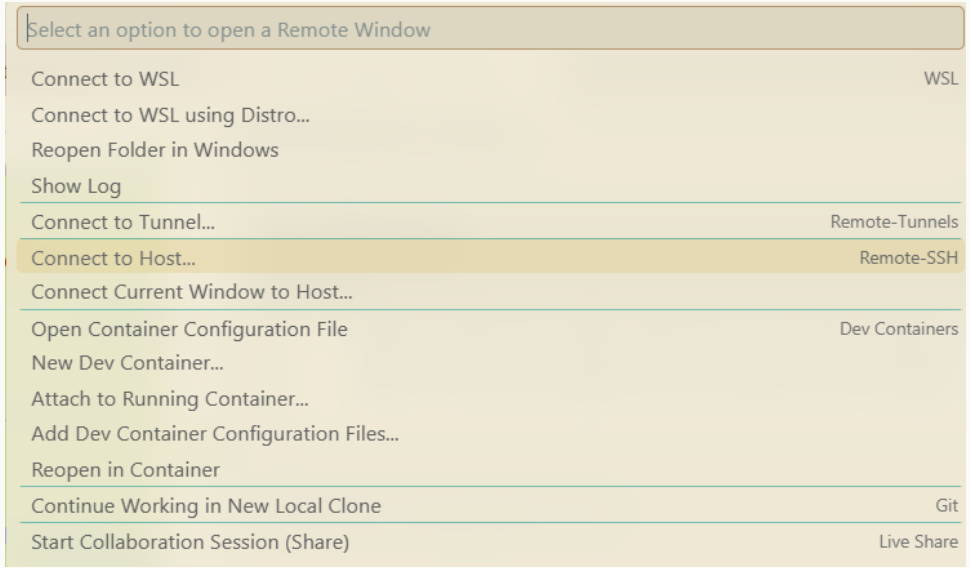

```
ms-vscode-remote.remote-ssh
```

STEP 7 · VS CODE · 2 OF 2

Connect to your host

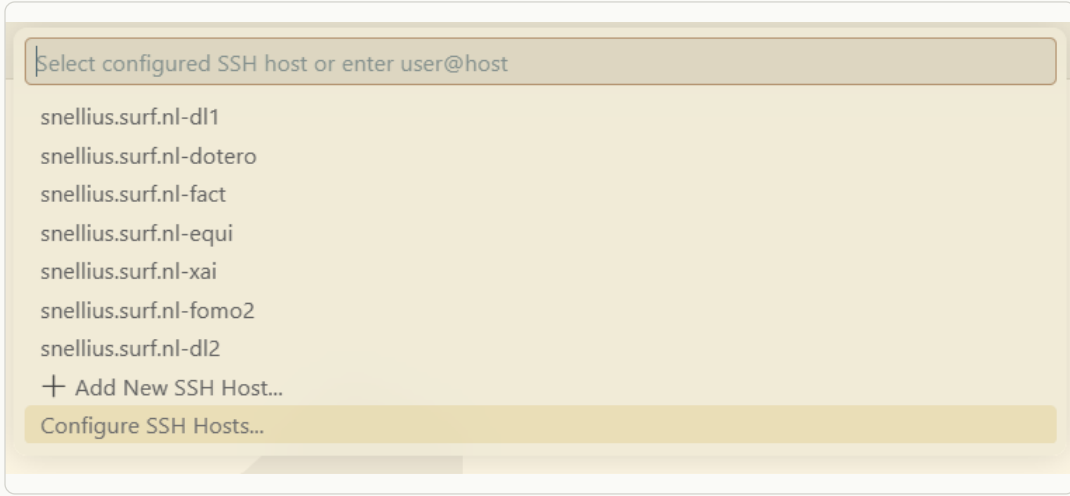
A Open the command palette

Press F1. Choose Remote-SSH: Connect to Host.



B Choose your host name

The name from your config appears in the list. Pick it, and a new VS Code window opens on Snellius.

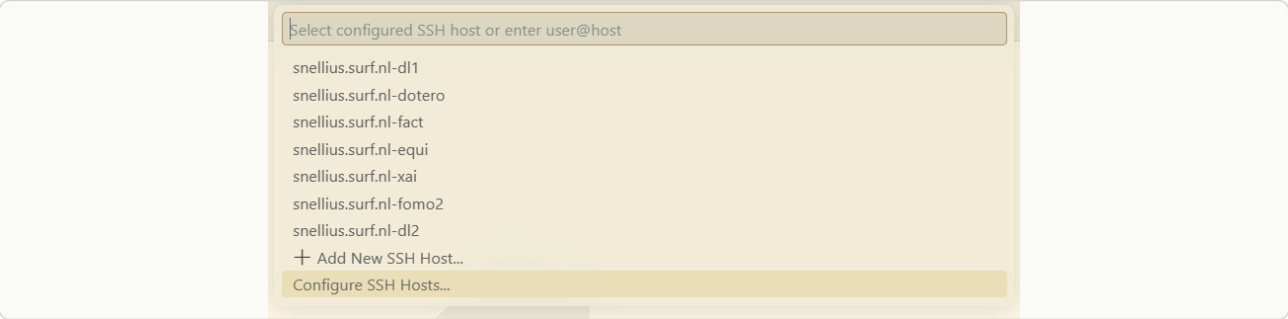


STEP 7 · VS CODE · SHORTCUT

Add your account from VS Code

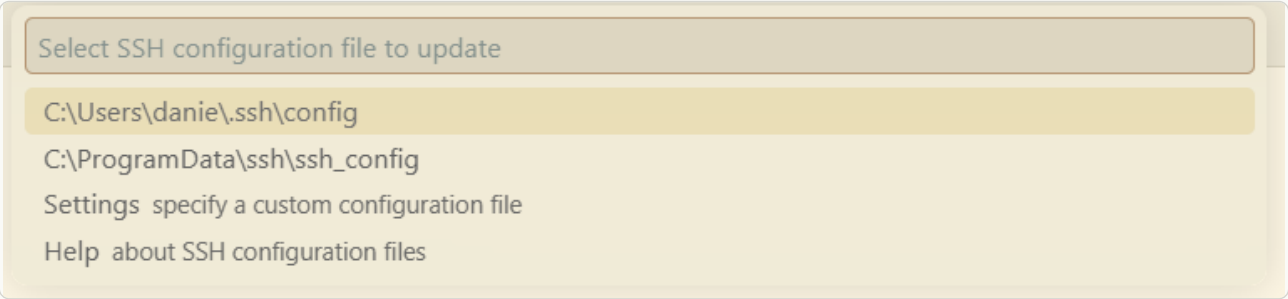
1 Configure SSH Hosts

Last option in the host list. Choose it to set up your Snellius account.



2 Pick your config file

Choose your user SSH config, then add the Host block and save.



STEP 7

Connect from a terminal

```
ssh <name>
```

> **<name>** is the Host name you added to your SSH config.

First time: type yes to accept the host fingerprint. No passphrase prompt if the key is in your agent.

Login nodes are shared. Edit and submit jobs there; run heavy work through sbatch or srun.

Troubleshooting

Symptom	Fix
Permission denied (publickey)	Not registered yet, or wrong User or IdentityFile. Run ssh -v <name>.
Cannot connect to the authentication agent	Windows: Start-Service ssh-agent as admin. macOS and Linux: eval "$(ssh-agent -s)".
Passphrase asked every time	Check ssh-add -l and keep AddKeysToAgent yes. Windows: set the agent service to Automatic.
Bad configuration option: usekeychain	Remove the UseKeychain line on Linux and Windows.
VS Code fails, terminal works	On Windows, set remote.SSH.path to the Windows ssh.exe, not the one from Git.

FIX FOR: UNPROTECTED PRIVATE KEY FILE

Fixing key permissions

macOS / Linux

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/id_ed25519_<key_suffix> ~/.ssh/config
chmod 644 ~/.ssh/id_ed25519_<key_suffix>.pub
```

Windows

```
$key = "$env:USERPROFILE\.ssh\id_ed25519_<key_suffix>"
icacls $key /inheritance:r
icacls $key /grant:r "$($env:USERNAME):(R)"
```

CHEAT SHEET

Quick reference

CREATE	`ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519_<key_suffix>`
LOAD	`ssh-add ~/.ssh/id_ed25519_<key_suffix>`
REGISTER	`cat ~/.ssh/id_ed25519_<key_suffix>.pub`
CONNECT	`ssh <name>`

macOS: add --apple-use-keychain to ssh-add. Windows: use $env:USERPROFILE\.ssh\ in place of ~/.ssh/.